

a large investor. For small investors, there are low-cost index products for value strategies in equity, fixed income, and currency markets as well. The pervasiveness of value across many different asset classes turns out to be something an asset owner should exploit in factor investing, which we come back to in chapter 14.

3.7. MOMENTUM

Another standard investment factor is momentum. This burst onto the academic scene with Jegadeesh and Titman (1993) in the same year that Fama and French were capturing size and value factors.³⁰ Industry professionals like Richard Driehaus, a star mutual fund manager, had already been practicing momentum for decades.³¹ Jegadeesh and Titman (1993) noted that Value Line, a vendor of financial data, has been providing price momentum signals in its publications since the 1980s.

Momentum Investing

Momentum is the strategy of buying stocks that have gone up over the past six (or so) months (winners) and shorting stocks with the lowest returns over the same period (losers). The momentum effect refers to the phenomenon that winner stocks continue to win and losers continue to lose. We call the momentum factor *WML*, for past winners minus past losers. (It is also called *UMD*, for stocks that have gone up minus stocks that have gone down.) The momentum strategy, like size and value, is a cross-sectional strategy, meaning that it compares one group of stocks (winners) against another group of stocks (losers) in the cross section, rather than looking at a single stock over time. Winners and losers are always relative—stocks win or lose relative to each other, and the market as a whole can go up or down.

Momentum returns blow size and value out of the water. Figure 7.7, which plots cumulated returns from January 1965 to December 2011, for *SMB*, *HML*, and *WML* speaks for itself. The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on size and value. Momentum is also observed in every asset class: we observe it in international equities, commodities, government bonds, corporate bonds, industries and sectors, and real estate.³² In commodities, momentum is synonymous with commodities

³⁰ Momentum had appeared in the literature with Levy (1967) but was ignored until Jegadeesh and Titman's (1993) work.

³¹ As recounted by Schwager (1992).

³² See Asness, Moskowitz, and Pedersen (2012) for momentum in equities, government bonds, currencies, and commodities. The standard momentum effect based on past returns is weak in Japanese equities, but versions of momentum do work in Japan; see Chaves (2012). For momentum in corporate bonds and real estate, see Jostova et al. (2013) and Marcato and Key (2005), respectively. Menkhoff et al. (2012b) is a detailed look at momentum in currencies.